



LABORATOIRE ONDES
et MILIEUX COMPLEXES



WaveLab

User Manual

Acquisition & Interpretation Interfaces

LOMC— Wave Dynamics Research Group

Version 1.0 — 2025

Audience: New laboratory users — no coding required

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1 Overview

WaveLab is a two-application software suite developed for experimental fluid mechanics research. It covers the complete data lifecycle from hardware acquisition to signal interpretation.

Element	Description
WaveLab Acquisition	Detects DAQ hardware, selects channels, records raw sensor data, and applies calibration.
WaveLab Interpretation	Loads calibrated CSV files, decomposes wave signals, and computes spectral metrics.

1.1 System Requirements

Element	Description
Operating System	Windows 10 or 11 (64-bit)
RAM	8 GB minimum
Python	3.9 or higher — python.org/downloads
USB Ports	One free USB port per MCC or NI acquisition card
Network	Required only for Kistler LabAmp devices (TCP/IP)

1.2 Supported Hardware

Element	Description
MCC USB-1808X	8-channel, 18-bit USB DAQ card — requires MCC DAQ Software + pip install mcculw
National Instruments	Any NI DAQ device supported by NI-DAQmx — requires NI-DAQmx driver + pip install nidaqmx
Kistler LabAmp	Network-connected amplifier via TCP/IP — enter IP address manually in the interface

☐ **Note:** If you do not have physical hardware, enable Simulation Mode for the MCC card. The software will generate realistic synthetic sine-wave signals so you can explore all features.

2 Acquisition Interface — WaveLab

The Acquisition interface is organised as a left-side navigation menu with five pages. Work through them from top to bottom for each experiment.

Element	Description
1. Device Detection	Scan for connected DAQ hardware and select which devices to use.
2. Channel Selection	Choose which analog input channels to record.
3. Acquisition	Set frequency and duration, then launch the real-time recording.
4. File Calibration	Convert raw voltage data to physical units (bar, N, m/s ² , ...).
5. Visualisation	Plot the calibrated data and inspect time-domain signals.

2.1 Page 1 — Device Detection

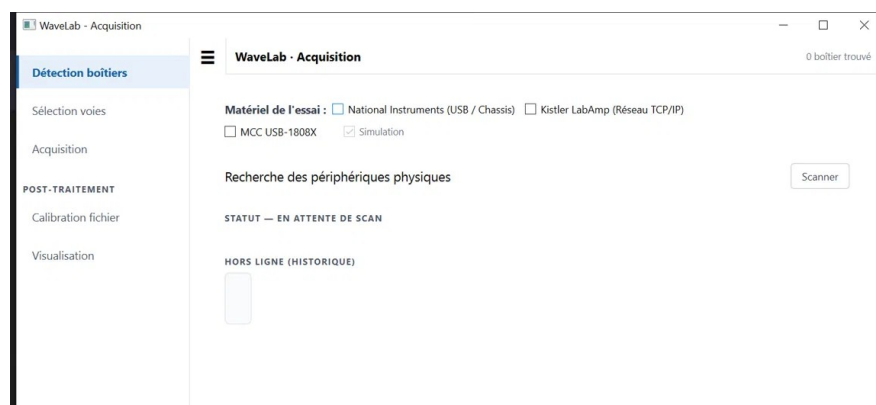
This page scans for DAQ hardware connected to the computer. You must complete this step before selecting channels.

Hardware selection

At the top of the page, check the box next to every hardware type you want to use:

- National Instruments (USB / Chassis) — for any NI DAQ device
- Kistler LabAmp (Network TCP/IP) — enter the device IP address in the text box that appears
- MCC USB-1808X — tick Simulation if no physical card is connected

⚠ Warning: Check at least one hardware type before clicking Scanner. If none are checked, the scan will not start.

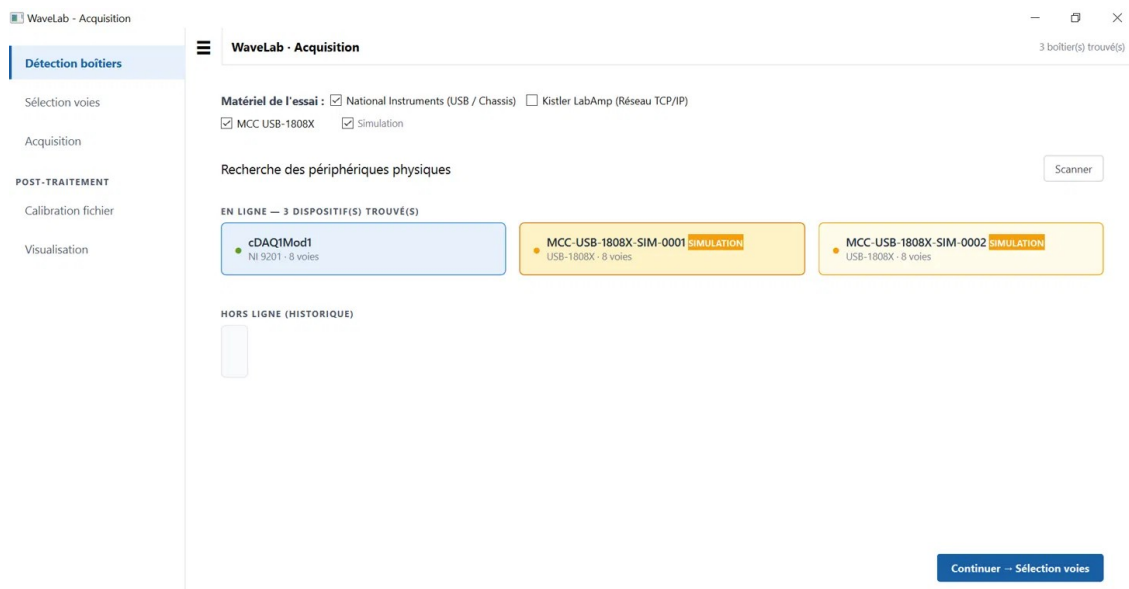


Running the scan

1. Connect all hardware to the computer (USB cables, network cables).
2. Select the hardware types you want to use by ticking the checkboxes.
3. For Kistler devices, type the IP address in the field that appears (e.g., 169.254.77.238).
4. For MCC without a physical card, tick Simulation next to MCC USB-1808X.
5. Click Scanner.
6. Detected devices appear in the ONLINE section as coloured cards. Simulated devices are shown with a orange SIMULATION badge.
7. Click Continue → Channel Selection to proceed.

Understanding the device cards

Element	Description
Green dot •	Device is physically connected and responding.
orange card (SIMULATION)	Device is running in simulation mode — no hardware needed.
Grey card (OFFLINE)	Device was previously used but is not currently reachable.



✓ **Tip:** If a device you expect does not appear, check the USB cable, verify the driver is installed (see the Installation Guide), and try scanning again.

2.2 Page 2 — Channel Selection

Select the analog input channels you want to record. Each channel corresponds to one physical sensor connected to the DAQ card.

How to select channels

8. The page shows one section per detected device.
9. Click on a channel button (e.g., ai0, ai1...) to toggle it on or off.
10. Selected channels are highlighted in blue.
11. Select as many channels as needed — at least one is required.
12. Click Continue → Acquisition to proceed.

Note: Channel names follow the format deviceName/aiN where N is the channel index (0–7 for MCC USB-1808X). These names appear as column headers in the exported CSV.

WaveLab - Acquisition

Détection boîtiers

Sélection voies

Acquisition

POST-TRAITEMENT

Calibration fichier

Visualisation

WaveLab · Sélection voies

3 boîtier(s) trouvé(s)

cDAQ1Mod1 - 8 voies

cDAQ1Mod1/ai0	cDAQ1Mod1/ai1	cDAQ1Mod1/ai2
cDAQ1Mod1/ai3	cDAQ1Mod1/ai4	cDAQ1Mod1/ai5
cDAQ1Mod1/ai6	cDAQ1Mod1/ai7	

MCC-USB-1808X-SIM-0001 - 8 voies

MCC-USB-1808X-SIM-0001/ai0	MCC-USB-1808X-SIM-0001/ai1	MCC-USB-1808X-SIM-0001/ai2
MCC-USB-1808X-SIM-0001/ai3	MCC-USB-1808X-SIM-0001/ai4	MCC-USB-1808X-SIM-0001/ai5
MCC-USB-1808X-SIM-0001/ai6	MCC-USB-1808X-SIM-0001/ai7	

Retour

Continuer → Acquisition

2.3 Page 3 — Acquisition

Configure the acquisition parameters and start recording. Data is streamed to a CSV file in real time while a live graph is updated on screen.

Acquisition mode

Choose how the acquisition will stop:

Element	Description
By Duration	Recording stops automatically after the specified number of seconds.
By Points	Recording stops after the specified number of samples per channel.

Setting the parameters

Element	Description
Duration (s)	Total recording time in seconds. Range: 1 – 3 600 s.
Sampling Frequency (Hz)	Number of samples per second per channel. Range: 1 – 150 000 Hz.
Number of Points	Visible only in By Points mode. Total samples to collect per channel.

WaveLab - Acquisition

Détection boîtiers

Sélection voies

Acquisition

POST-TRAITEMENT

Calibration fichier

Visualisation

WaveLab - Acquisition

3 boîtier(s) trouvé(s)

PARAMÈTRES DE L'ESSAI

DURÉE (S)

50

FRÉQUENCE (HZ)

500

VOIES ACTIVES

5 sélectionnées

Nombre de points enregistrés = Durée × Fréquence (ex: 10 s × 1000 Hz = 10 000 pts/voie)

En attente

0.0 s / 10 s

Lancer l'essai

0.8

0.6

0.4

0.2

0.1

0.2

0.3

0.4

0.5

0.6

0.7

0.8

0.9

Launching and stopping the acquisition

13. Set Duration and Frequency
14. Click Launch Test. The button changes to Stop, a progress bar appears, and the graph starts updating.
15. The status indicator shows In progress in blue.
16. The acquisition stops automatically when the duration or point count is reached.
17. To stop early, click Stop at any time.
18. When finished, a green banner appears with two actions: Calibrate this file or Save as...

Acquisition terminée avec succès
9.9 s · 500 Hz · 5 voies · 4948 pts

[Calibrer ce fichier](#)[Enregistrer sous...](#)

Live graph

Each selected channel is plotted in a different colour. The graph updates in real time. High-frequency recordings (> 5 000 Hz) are automatically down-sampled for display while the full data is always written to the CSV file.

✓ **Tip:** Click Calibrate this file in the green banner to go directly to calibration with the current file pre-loaded.

WaveLab - Acquisition

Détection boîtiers

Sélection voies

Acquisition

POST-TRAITEMENT

Calibration fichier

Visualisation

WaveLab · Acquisition

PARAMÈTRES DE L'ESSAI

DURÉE (S)

50

FRÉQUENCE (HZ)

500

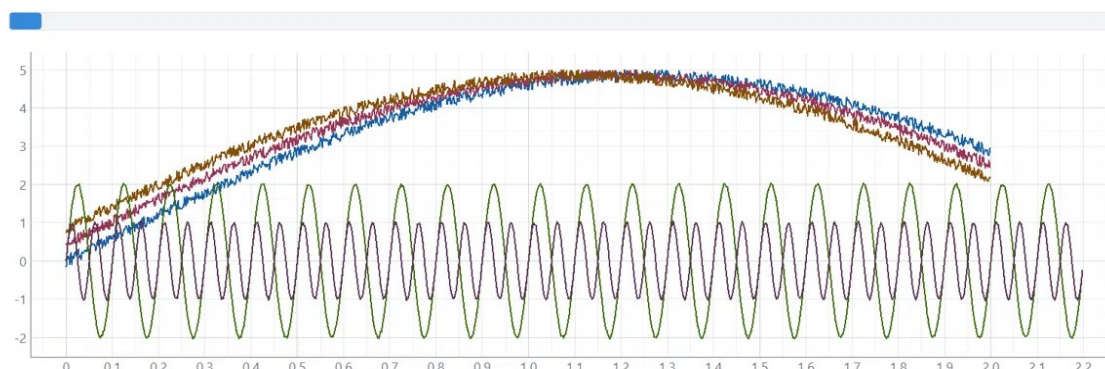
VOIES ACTIVES

5 sélectionnées

Nombre de points enregistrés = Durée × Fréquence (ex: 10 s × 1000 Hz = 10 000 pts/voie)

En cours 2.0 s / 50 s

Arrêter



2.4 Page 4 — File Calibration

Calibration converts the raw voltage values (V) recorded by the DAQ card into physical units (bar, N, m/s², °C, m/s) using a two-parameter linear formula derived from the sensor datasheet.

Loading a file

You can reach this page in two ways:

- Directly after acquisition: click Calibrate this file in the green success banner. The file is loaded automatically.
- Manually: click File Calibration in the left menu, then drag-and-drop or click the drop zone to browse for any CSV file.

Configuring each channel

Once a file is loaded, one row appears per data channel. For each row:

19. Select the sensor Type from the drop-down (Pressure, Force, Acceleration, Velocity, Probe, Raw).
20. The parameter labels update to show the physically meaningful names from the sensor datasheet.
21. Enter the two calibration values from your sensor certificate or datasheet.
22. The formula preview on the right updates in real time.

WaveLab - Acquisition

Détection boîtiers

Sélection voies

Acquisition

POST-TRAITEMENT

Calibration fichier

Visualisation

WaveLab · Calibration

3 boîtier(s) trouvé(s)

Fichier source

donnees_fusionnees.csv

4948 lignes · 2 colonnes · Données directes

Parcourir...

Définition des capteurs par voie

VOIE TYPE

UNITÉ A (PENTE) C (OFFSET) FORMULE

Ch1

Pression

Pression

Force

Accélération

Vitesse

Sonde

Brut

bar

Pleine Échelle PE (bar)

10,0000

Tension zéro V_zéro (V)

0,0000

$y = PE \times (V - V_{ref}) / (10 - V_{ref}) \text{ [bar]}$

Ch2

bar

Pleine Échelle PE (bar)

10,0000

Tension zéro V_zéro (V)

0,0000

$y = PE \times (V - V_{ref}) / (10 - V_{ref}) \text{ [bar]}$

Visualiser calibré

Enregistrer fichier calibré...

Calibration formulas by sensor type

Sensor Type	Parameter 1	Parameter 2	Formula	Output Unit
Pressure	Full Scale PE (bar)	Zero Voltage V_zero (V)	$PE \times (V - V_zero) / (10 - V_zero)$	bar
Force	Nominal Capacity CN (N)	Sensitivity S (mV/V)	$CN \times V / (S/1000 \times 10)$	N
Acceleration	Sensitivity S (mV/g)	Bias Voltage V_bias (V)	$(V - V_bias) \times 1000/S \times 9.81$	m/s ²
Velocity	Sensitivity S (mV/(m/s))	Offset V_offset (V)	$(V - V_offset) / (S/1000)$	m/s
Probe (Temp)	Conditioner Gain G (°C/V)	Zero Temp T_zero (°C)	$G \times V + T_zero$	°C
Raw	Gain a	Offset b	$a \times V + b$	V

☐ **Note:** Parameter values come from the sensor datasheet or calibration certificate. If in doubt, contact the sensor manufacturer. For a raw voltage channel with no conversion, leave Type = Raw with a = 1 and b = 0.

Saving the calibrated file

23. Click Save calibrated file....
24. Choose a location and filename in the dialog.
25. The calibrated CSV is saved with renamed columns indicating the unit (e.g., ai0(bar), ai1(N)).

✓ **Tip:** The original raw file is never modified. You can re-calibrate the same raw file as many times as needed with different parameters.

2.5 Page 5 — Visualisation

Plot a calibrated CSV file to visually inspect the time-domain signals before running interpretation algorithms.

Loading and plotting

- Click Visualise calibrated in the calibration success banner, or navigate to Visualisation in the left menu.
- A file browser opens — select the calibrated CSV file.
- Each data column is plotted as a separate curve with a unique colour.
- The legend shows the channel name and its unit (e.g., ai0 [bar]).

Interface elements

Element	Description
Channel selector (top right)	Show one channel at a time or all channels simultaneously.
Tabs Brut / Calibr�	Switch between the raw voltage view and the calibrated physical-unit view.
Metric boxes	Display quick statistics: max amplitude, peak period, global maximum, and duration.
Plot area	Interactive graph — scroll to zoom, click and drag to pan.



3 Interpretation Interface

The Interpretation interface processes calibrated CSV files produced by WaveLab Acquisition. It applies wave decomposition algorithms and spectral analysis methods to extract physical wave parameters.

3.1 Interface Overview

Element	Description
Left panel	File loading, method selection, parameter form, and the Run button.
Right panel	Interactive graph, zoom button, and a log console.
Column bar	Checkboxes to select which data columns (probes) to include in the calculation.
? button	Opens the built-in tutorial at any time.

The screenshot shows the 'Interface de Traitement de Données' window. The left panel contains a file path input field with 'D:/HP/Test_2016.csv' and a 'Parcourrir' button. Below this is a row of checkboxes for data columns: '# t[s]', 'S1 [V]', 'S2 [V]', 'S3 [V]', 'S4 [V]', 'S5 [V]', 'S6 [V]', 'S7 [V]', 'S8 [V]', 'S9 [V]', 'S10 [V]', 'S11 [V]', and 'Unn'. The 'S5 [V]' and 'S6 [V]' checkboxes are checked. Below the checkboxes is a horizontal scrollbar. Further down, there are three dropdown menus: 'Decomposition_EldrupAnderson' (highlighted in green), 'Conjuguee', and 'Imagerie'. Below these are radio buttons for 'Irregular:' with 'YES' and 'NO' options, where 'NO' is selected. There are four input fields: 'ProbeSpacing' with the value '0,0,0,3,0,7,1,2', 'Depth' with the placeholder 'Valeur pour Depth (m)', 'Fréquence' with the placeholder 'Valeur pour Fréquence (Hz)', and 'order' with the placeholder 'Valeur pour order'. Below these is a section for 'Paramètres Additionnels' with a 'Gravity' dropdown menu. At the bottom of the left panel is a large blue 'Spectre' button. The right panel features a graph with a y-axis from 0 to 1.0 and an x-axis from 0 to 0.8. Below the graph is a blue 'Zoomer graph' button. At the bottom of the right panel is a large blue 'Visualisation' button.

3.2 Step-by-Step Workflow

Step 1 — Load a data file

30. Click Browse (Parcourir) next to the file path field.
31. For signal analysis methods: select a CSV file (e.g., the calibrated output from WaveLab Acquisition).
32. For Imagerie method: select a calibration image (PNG, JPG, BMP, or TIF).
33. The column names appear as checkboxes in the bar below the file path.

Step 2 — Select data columns

Tick the checkboxes corresponding to the probe/sensor columns you want to analyse. The time column is always selected and cannot be deselected.

✓ **Tip:** Select only the columns relevant to your experiment. Including unnecessary columns does not improve accuracy and may slow the calculation.

Step 3 — Choose the analysis method

Three drop-down menus at the top of the left panel select the method:

Element	Description
Menu 1 — Decomposition	WaveProbeDecomposition Decomposition_LiuHuang Decomposition_EldrupAnderson
Menu 2 — Dsp	Standard Conjuguee
Menu 3 — Image	Imagerie (image-based free-surface detection)

Step 4 — Fill in the parameters

The parameter form updates automatically when you change the method. The required fields depend on the selected algorithm:

Element	Description
ProbeSpacing (m)	Comma-separated list of probe positions measured from the first probe (e.g., 0.0, 0.3, 0.7, 1.2).
Depth (m)	Water depth at the probe location.
Frequency (Hz)	Angular or cyclic frequency of the incident wave (used in LiuHuang and EldrupAnderson methods).
order	Harmonic order for non-linear decomposition (integer, typically 2)

or 3).

Step 5 — Optional additional parameters

A secondary drop-down labelled Additional Parameters offers optional settings:

Element	Description
Gravity	Override the gravitational constant g (default: 9.81 m/s^2).
StartCutIndex	Index of the first sample to include (removes a transient at the start of the signal).
Integration	Set the frequency band $[f_{\min}, f_{\max}]$ for spectral energy integration.

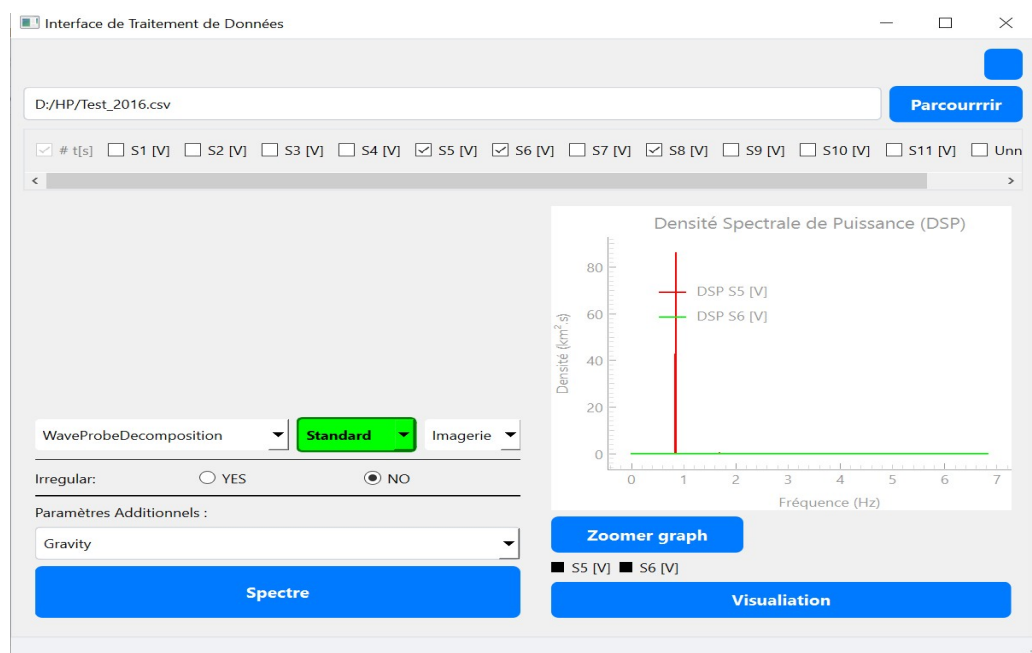
Step 6 — Set the irregular wave flag

Irregular: YES — use for random or irregular sea states (multiple frequency components).

Irregular: NO — use for monochromatic (single-frequency) regular waves.

Step 7 — Run the analysis

34. Click Spectre to launch the computation.
35. For decomposition methods: a pop-up shows the incident amplitude (A_i) and reflected amplitude (A_r).
36. For spectral methods: the Power Spectral Density (PSD) is plotted on the right panel with frequency on the x-axis (Hz) and energy density on the y-axis ($\text{m}^2 \cdot \text{s}$).
37. Click Zoom Graph to open the plot in a larger window for inspection.
38. Clic visualisation to plot the raw time-domain signal for all selected channels.



3.3 Imagerie Mode — Free-Surface Detection

The Imagerie mode uses computer vision to detect and reconstruct the free water surface from a sequence of images. It requires a calibration pattern (chequered mire) to correct lens distortion.

Required inputs

- A calibration image of the chequered mire (PNG, JPG, or TIF).
- A folder of experiment images.
- Pattern_X and Pattern_Y: number of inner corners of the chequered pattern.
- Sigma_Flatfield: smoothing parameter for flat-field correction.

Processing sequence

39. Select Imagerie from Menu 3. The graph panel is replaced by image processing buttons.
40. Click Browse and load the calibration image.
41. Fill in Pattern_X, Pattern_Y, and Sigma_Flatfield.
42. Step 1 — Click Windowing & Mire Detection: the software detects the corners of the calibration pattern.
43. Step 2 — Click Radial Distortion & Perspective: corrects lens distortion and perspective. A corrected preview is shown.
44. Step 3 — Click Image Correction: select the folder of experiment images. Corrected images are saved.
45. Step 4 — Click Free Surface Detection: extracts the water surface elevation from each corrected image.

☐ **Note:** All intermediate results and corrected images are saved in the folder you selected in Step 3.

4 Complete Experiment Workflow

The following diagram summarises the full data flow from hardware setup to final results.

1 Connect Hardware

Plug in USB-1808X, NI, or Kistler devices. Ensure drivers are installed (see Installation Guide).

2 Launch WaveLab Acquisition

Run InterfaceAcquisition.py.

3 Detect Devices

Page 1 — Select hardware types and click Scanner.

4 Select Channels

Page 2 — Click the channels you want to record.

5 Record Data

Page 3 — Set frequency and duration, then click Launch Test.

6 Calibrate

Page 4 — Assign sensor types and calibration parameters. Click Save calibrated file.

7 Launch WaveLab Interpretation

Run interface.py.

8 Load Calibrated File

Click Browse and select the calibrated CSV.

9 Select Method & Parameters

Choose the decomposition or spectral method and fill in the form.

10 Run & Export Results

Click Spectre. Read results from the pop-up or the PSD graph.

5 Troubleshooting

Problem	Solution
0 devices detected after scan	Check USB cables. Verify that the driver (MCC DAQ Software or NI-DAQmx) is installed. Open InstaCal or NI MAX to confirm the device is seen by the operating system.
MCC card detected in InstaCal but not in WaveLab	Run WaveLab as Administrator (right-click → Run as administrator).
Acquisition stops immediately	Check that the ScanThread is passing scan_mcc and mode_simulation_mcc parameters correctly. Verify mcculw is installed: pip install mcculw.
PermissionError on temp_mcc.csv	A previous acquisition did not close the file. Restart WaveLab. The file will be released.
No file source loaded on Calibration page	If you arrived from the Acquisition page, ensure the acquisition completed and a success banner was shown before clicking Calibrate this file.
ModuleNotFoundError: mcculw	Run: pip install mcculw and ensure MCC DAQ Software is installed first.
ModuleNotFoundError: nidaqmx	Run: pip install nidaqmx and ensure NI-DAQmx driver is installed.
Interface does not launch	Run: pip install PyQt6 pyqtgraph then retry.
Decomposition result seems incorrect	Verify that ProbeSpacing values match the physical distances between probes (in metres). Check that the correct Irregular setting is selected.
Graph is blank after loading a file	Confirm that at least one column checkbox is ticked in the column bar and that the time column is present in the CSV.

6 Installation Quick Reference

Perform these steps in order on every new computer that will run WaveLab.

6.1 Python dependencies (all users)

```
pip install -r requirements.txt
```

The requirements.txt file installs: PyQt6, pyqtgraph, pandas, numpy, mcculw, nidaqmx.

6.2 Hardware-specific drivers

Element	Description
MCC USB-1808X	Download MCC DAQ Software from digilent.com/reference/software/mccdaq-cd/start — disconnect the card before installing, reconnect after.
National Instruments	Download NI-DAQmx from ni.com/fr-fr/support/downloads/drivers/download.ni-daq-mx.html — restart the PC after installation.
Kistler LabAmp	No driver required. Connect to the same local network and enter the IP address in WaveLab.

6.3 Verify the installation

```
python verifier_installation.py
```

The script checks Python version, installed libraries, and detected hardware. All items should show [OK]. If any item shows [XX], follow the instruction printed next to it.

✓ **Tip:** For any issue not covered by this manual, consult the MCC forum at forum.digilent.com or the NI community at forums.ni.com.